**Final Self Critique**

In practice, physicians would typically be combining multiple low-resolution MRI images into a single high-resolution one using these reconstruction methods: using multiple images or even an entire stack would introduce computational constraints as the filesize increases.

Even our most technically advanced reconstruction  model, SSDiffRecon, doesn't match increased accuracy that the researchers reported, likely due to our smaller training set and changes in implementation. Giving the model a more difficult objective to solve (with a smaller k-space mask and a stricter data consistency requirement) makes it more difficult for the model to make changes from the input. Future work may explore using larger datasets that are optimized for MRI reconstruction, such as fastMRI.

Across all methods and both datasets, the most consistent finding was that zero-filled FFT provided a strong baseline, one that kernel reconstruction rarely surpassed. This pattern held for both MRI and natural images, which shows that the structural assumptions kernel basis functions use may be too restrictive when compared to the simplicity of zero-filled FFT. The best strategy throughout was to treat zero-filled FFT as the starting point and apply a learned refinement on top of it, with the residual CNN combination achieving the best multi-slice MRI results at 33.20 dB PSNR.

Taking the project from MRI to natural RGB images using the Oxford-IIIT Pet Dataset showed that these reconstruction patterns generalize across image types and not just MRIs. While the preferred kernel type differed, the relative ordering of methods remained consistent, with zero-filled FFT plus residual CNN doing better than the kernel-based in both types of images.

The strongest absolute results came from multi-frame methods, where MFSR-GAN reached 38.66 dB PSNR which leverages temporal context across adjacent frames. A natural next step would be applying multi-frame reconstruction directly to MRI stacks, where adjacent slices provide similar adjacent contexts. With access to a larger benchmark dataset, and with the diffusion model fully functional, future work could better evaluate where each method sits relative to SOTA reconstruction methods.

In the future, we would also like to explore more efficient hyperparameter tuning methods. Parameter tuning was computationally expensive, requiring us to restrict the search space or the training set of images. If we had more compute, exploring better hyperparameter tuning methods or making our method more robust would be the first thing we would explore.